

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2016

PHYSICS

9188/3

2014/2017

Back credit - 6x

B.C for numerical values

1 (a) (i) zero-error reading = 0.05 cm 2 ignore absolute error in the readings. B1
 = (4.48 ± 0.01) cm ~~Both correct~~ B1

(ii) 1. diameter = $(4.48 - 0.05) \pm 0.01$ cm
 = (4.43 ± 0.01) cm A1

2. Volume = $\pi r^2 l = \frac{\pi d^2 l}{4}$
 = $\frac{\pi}{4} (4.43^2) (25.0)$ cm³ Accept 3.s.f. C1
 = 385.33 cm³ ignore units C1

error in
 repeat ± 0.005 in radius
 if wrong
 $r = 2.215$
 out of range error
 DORE

$\Delta V = \left(\frac{2\Delta d}{d} + \frac{\Delta l}{l} \right) (\text{Volume})$ Accept e.c.f. in the error of the diameter. B1
 = $\left(\frac{2 \times 0.01}{4.43} + \frac{1}{250} \right) (385.33)$ cm³ C1
 = 3.281 cm³

$\therefore V = (385 \pm 3)$ cm³ In precision - values compared to their mean A1
Degree of closeness Accuracy - values compared to True Value

absoluteness of precision
 to random error
 - precision is separate

(b) (i) precision is closeness of values to mean value while accuracy is closeness of mean value to true value. B1
 (ii) Set A: precise but not accurate A/W B1
 Set B: accurate but not precise meet/intersect/overlap B1

(c) (i) (States that whenever) two waves are travelling in the same region the total displacement at (any point) is equal to the vector sum of their individual displacements at that time resultant displacement is vector/Phasor sum B1
accept sum of Amplitudes

(ii) $V = f\lambda$

$\therefore \lambda = \frac{330}{1320}$

= 0.25 m - ignore units C1

i.e distance between A and Z = $\frac{\lambda D}{a}$

$$= \frac{0.25 \times 7}{1.5}$$

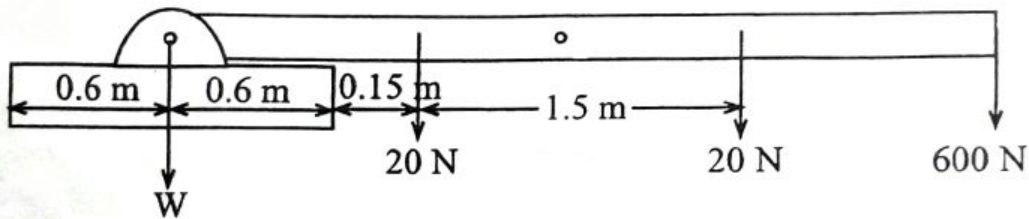
$$= 1.17 \text{ m}$$

C1

A1

- (d) (i) Product of the force and its perpendicular distance from pivot to its line of action)
- (ii)

B1



Clockwise moments = anticlockwise moments

B1 C1

$$W \times 0.6 = (20)(0.15) + (20)(1.65) + (2.4)(600)$$

C1

$$W = 2460 \text{ N}$$

A1

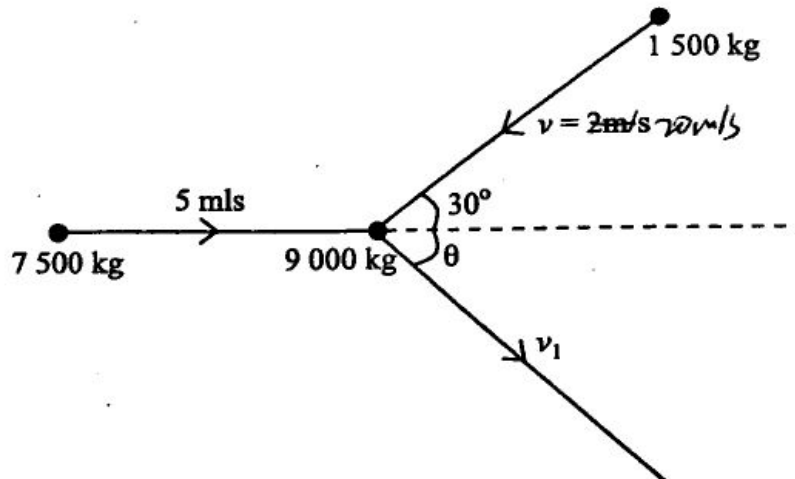
- 2 (a) (i) $Y_2 = 2.00 \times 10^{-3} \sin[(600)t + (20.0)x]$ B1
- (ii) - the waves in opposite directions *opposite dir* B1
 - they have the same amplitude/frequency and speed B1
- (iii) Resultant displacement $Y_R = Y_1 + Y_2$ *superf. princ. principle*
 $= 2.00 \text{ mm} \times 10^{-3} (\sin[(600)t - (20.0)x] + \sin[(600)t + (20.0)x])$ *$Y_R = 2 \sin(\omega t) \cos(kx)$ (3 marks) $\times 2$* C1
 $= 4.00 \times 10^{-3} \cos[(20.0)x] \sin[(600)t]$ A1
- (b) (i) $\frac{-2\pi x}{\lambda} = -(20.0)x$ C1
- $\therefore \lambda = \frac{\pi}{10} \text{ m}$ *correct* $2 = 0.314$ A1
- $V = \frac{dy}{dt}$ *correct* $V = \omega y$
- $= (2.00 \times 10^{-3})(600) \cos[(600)t - (20.0)x]$ C1
- $\therefore V_{\max} = 2.00 \times 10^{-3} \times 600 = 1200 \times 10^{-3} \text{ ms}^{-1}$ A1
- (c) To replenish lost energy due to damping. / *maintain Amplitude constant* B1
A/W

equation, terms
defined

- (a) - change of momentum
(A) unit = ~~Ns~~ Ns
@ unit is kgms⁻¹

B1
B1

- (b) (i)



Consider vertical component

$$0 = 1500 \times 20 \sin 30 - 9000V \sin \theta$$

C1

$$\Rightarrow V \sin \theta = 1.667 \quad \text{-----} \quad (1)$$

horizontal component

$$7500 \times 5 = 1500 \times 20 \times \cos 30 + 9000V \cos \theta$$

C1

$$\Rightarrow V \cos \theta = 1.2799 \quad \text{-----} \quad (2)$$

divide (1) by (2)

$$\frac{V \sin \theta}{V \cos \theta} = \frac{1.667}{1.2799}$$

C1

$$\theta = \tan^{-1} \left(\frac{1.667}{1.2799} \right)$$

$$= 52.5^\circ$$

A1

$$\therefore V = \frac{1.667}{\sin 52.5}$$

C1

$$= 2.10 \text{ ms/}$$

A1

28
11

- (c) - head tends to stay at the same place while body moves forward B1
 - hence head restraint pushes the head forward so that the whole body moves forward as one B1

- 4 (a) (i) work done per unit mass in bringing it from infinity to that point in a gravitational field *Accept (on a unit mass) except definition in point mass in symbols defined* B1
 (ii) force per unit mass *r - distance from centre of planet to point* B1
- (b) (i) 1. By convention, at infinity potential is zero, hence negative values are less than zero *A/W - gravitational force is attractive* B1
 2. its field is negligible *A/W* so that it does not distort the field under investigation B1

(ii) $E_k = \frac{1}{2} Mv^2$

$$\frac{MV^2}{R} = \frac{GM_E M}{R^2}$$

$$V^2 = \frac{GM_E}{R}$$

$$\therefore E_k = \frac{GM_E M}{2R}$$

$$\frac{1}{2} \left(\frac{MV^2}{R} \right) = \frac{1}{2} \left(\frac{GM_E M}{R} \right)$$

- (c) - satellite has both KE and PE *loss in PE > gain in KE* B1
 - the loss in potential energy is twice the gain of KE in orbit, hence all in all there is loss of energy *A/W* B1

— energy is lost as heat

— energy is lost as other forms instead of it being converted to K.E

$1 = \frac{F}{r}$

All attractive forces are associated with negative potential